SOCA

Sobaie Catchment Analysis

User Manual and Technical Guide

From a study area drawn on the map to catchments, sub-basins, morphometry, land use, soil groups and curve numbers – with a report, GIS files, map sheets and a ready-to-run HEC-HMS model. Every step, formula and data source is explained together with the publication it comes from.

Tool	SOCA – Sobaie Catchment Analysis, portable SOCA.1.0.exe
Author	Mohamed Abosobaie (Sobaie)
Document	User manual, methods, verification and references
Edition	November 2026 – for SOCA version 1.0

A reference tag such as R7 points to the list in Chapter 13; a Tool tag marks a convention chosen in SOCA rather than taken from a standard. All screenshots use synthetic example data – an invented mountain wadi of about 250 km² (2,050–3,330 m, land use and soils made up) crossed by a road – no real project or place.

Contents

1 What SOCA does

2 Getting started – access, screen, AI assistant

3 Step 1 – Study area and outlets

4 Step 2 – Sub-basins

5 Land use, soil and CN (automatic)

6 Step 3 – Advanced settings

7 Running – progress, area cap, offline

8 Map tab – layers and labels

9 Results tab

10 Layout tab – map sheets

11 Exports

12 Projects and privacy

13 References

A Appendix A – Methods and formulas

B Appendix B – Verification of SOCA

C Appendix C – Limitations and good practice

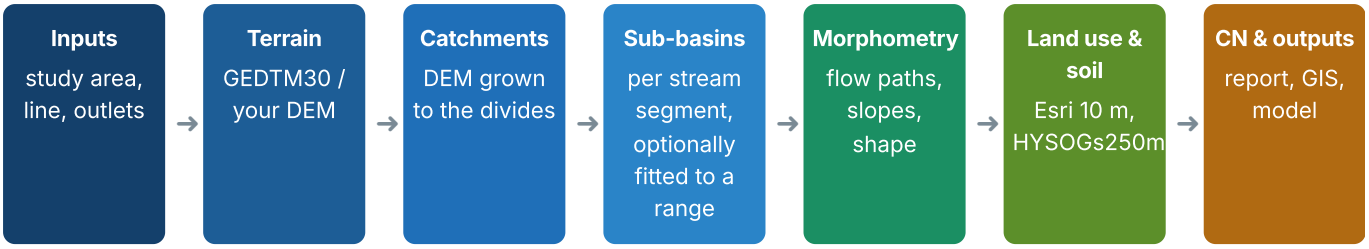
D Appendix D – Glossary

Quick start (5 minutes)

1. Drop your study area (KMZ, KML, GeoJSON or shapefile – a road corridor, a site boundary or a line across the wadis) into step **1**, or draw it with the tools on the left of the map. Add outlet points for culverts or dams if you have them. A second line further upstream adds sub-basin outlets where it crosses streams.
2. Check step **2 Sub-basins**: stream threshold, minimum and maximum sub-basin area, and whether to fit the sub-basins to the min – max range.
3. Step **3 Advanced settings** is optional: terrain data, area cap, and your own DEM or CN raster. Land use, soil groups and curve numbers (ARC II) need no input.
4. Press **Run**. SOCA downloads only the terrain, land use and soil data of your area, grows the DEM until no catchment touches its edge, and fills the Map, Results and Layout tabs.
5. Open **Export ▾** for the PDF / Word report, Excel tables, GIS files, rasters and the HEC-HMS model; the **Layout** tab exports map sheets as PDF or images. Everything is kept in the project folder.

1 What SOCA does

SOCA turns the first, repetitive part of every hydrology study – finding what drains to the site and describing each sub-basin – into one run. It is meant for road, rail, airport and urban drainage projects, dams and flood studies, where an engineer needs catchments, sub-basins and their parameters before any rainfall–runoff model can be built.



1.1 What you get

- **Catchments** of every stream entering your study area (or crossing your study line, or reaching your outlet points), complete up to the divide – the DEM is enlarged automatically until no catchment touches its edge.
- **Sub-basins** named consistently (B-1 … for one catchment, B-1-1 … for several), with **outlets** (O-1 … / O-1-1 …) and **reaches** joining them.
- **Morphometry** of every sub-basin: area, longest flow path, its drop, slope and 10–85 slope, mean land slope, main reach, SCS lag and T_c (lag / 0.6), Kirpich T_c for comparison, and the shape indices Ff, Re and Kc.
- **Land use, hydrologic soil group and curve number** per sub-basin (area-weighted, average antecedent runoff condition ARC II, used for the losses and the SCS lag; ARC I and III values are listed in the Excel tables for reference).
- **Outputs**: PDF and Word report, Excel tables, KMZ, shapefiles, GeoJSON, GeoTIFF rasters, map layouts (PDF / image) and a complete **HEC-HMS 4.14 project** that opens, shows the map and computes.

1.2 Running SOCA

SOCA is one portable file, **SOCA 1.0.exe** – no installation and no administrator rights. It needs Windows 10 or 11 (64-bit) with Microsoft Edge WebView2, which Windows already includes. The first start unpacks the program once into SOCA's own hidden folder (%LOCALAPPDATA%\SOCA, about 10 s); later starts take about 2 s. Older versions are removed automatically. Windows may warn that the file is from an unknown publisher (it is not code-signed): choose *More info → Run anyway*.

2 Getting started

2.0 Opening SOCA – access and messages

SOCA checks its access online every time it starts, so an **internet connection is needed at start**; it does not open offline. What you see depends on how the developer has set access:

- **Open access** – SOCA opens straight away, no password.
- **Password** – type the password you were given and press **Open SOCA**. It is remembered on this computer for **14 days**: until then SOCA opens without asking. It is asked again after 14 days, when the developer changes the password, or when open access is switched off.

While SOCA is open it re-checks every few hours; if access has ended you are asked to save, then SOCA closes. The start can also show:

- **Messages** from the developer – notices, and popups shown once per computer (several unseen ones together). **OK** becomes active after a short 3-second ring, so the message is read.
- **New version** – a message with **Later** and **Download**: Download saves the new SOCA in your Downloads folder and opens it; the old copy stays until you delete it. When the developer requires a newer version, an older copy does not open and offers only Download.
- **Too many attempts** (three wrong passwords within 15 minutes), **Daily limit for this computer reached**, or **This computer has been blocked** – wait, or contact the developer (contact line on the start screen).

On the first start SOCA shows the Terms of Use; scroll to the end, tick the box and press **Accept** (Cancel closes SOCA). They are shown again only when they change, and can be read any time from **Terms of use** in the top bar.

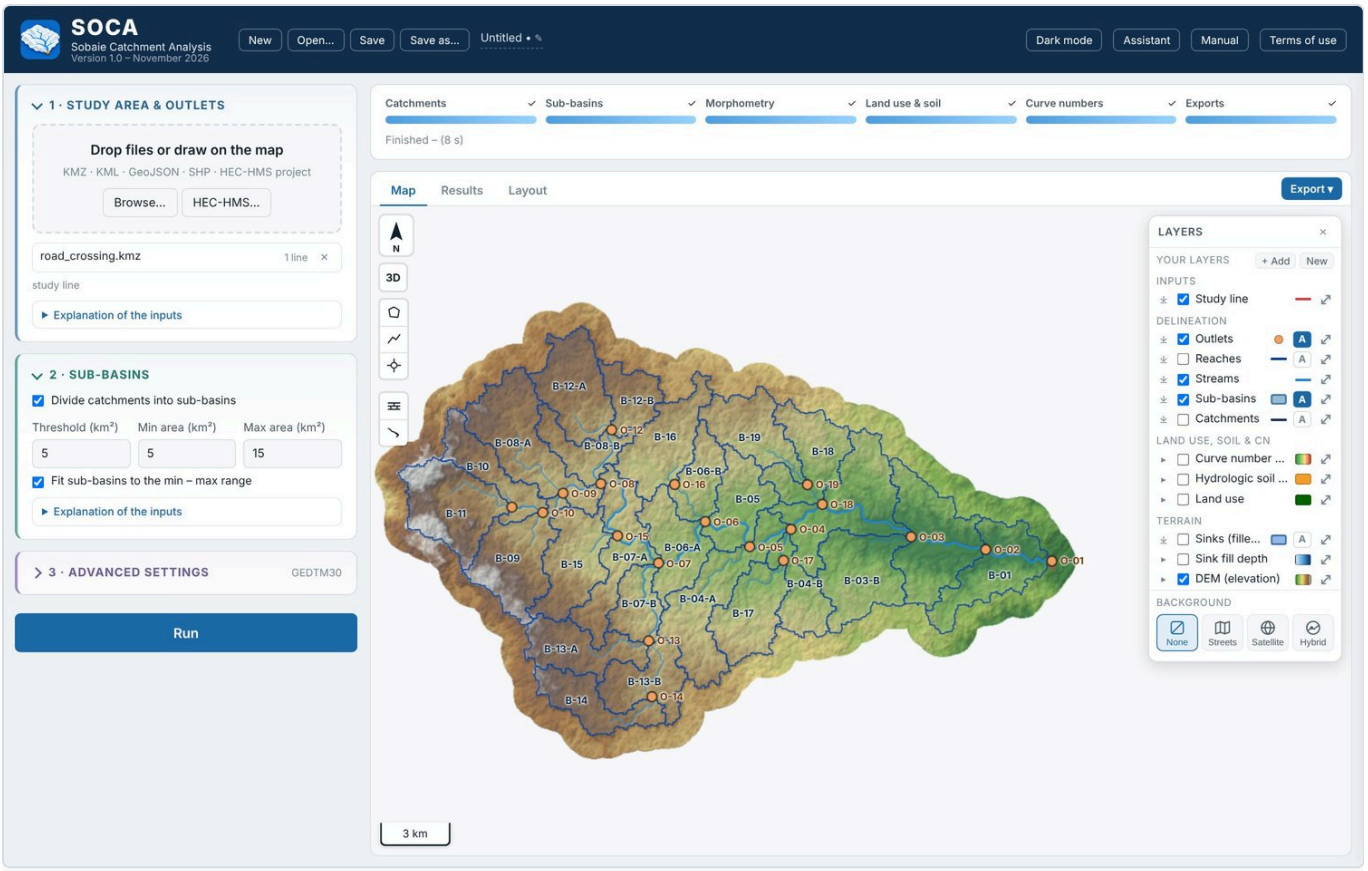


Figure 2.1 – The SOCA window after a run: a road crossing a mountain wadi, 26 sub-basins fitted to 5–15 km² (synthetic example terrain).

Area	What it does
Top bar	**New**, **Open...**, **Save**, **Save as...** projects (.soca); the project name (click to rename); **Dark mode**; **Assistant** (Chapter 2.1); **Manual**; **Terms of use**.
Left panel	The three input steps – each can be folded – with an *Explanation of the inputs* under each step, and the **Run** button (Ctrl + Enter). During a run it becomes **Stop**.
Progress bar	Six steps: Catchments, Sub-basins, Morphometry, Land use & soil, Curve numbers, Exports; the current action and the elapsed time.
Map / Results / Layout	The map with all layers; the result tables; printable map sheets. **Export ▾** appears after a run.

2.1 AI assistant

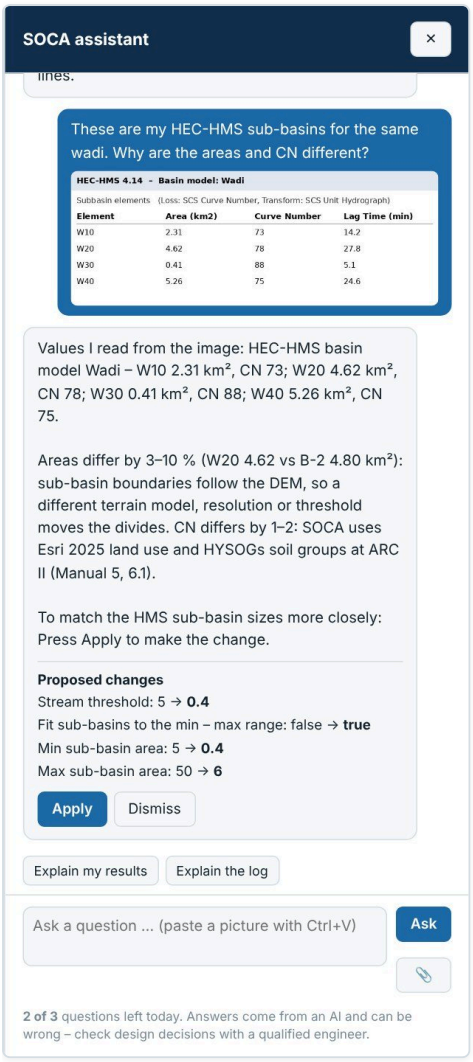


Figure 2.2 – The SOCA assistant: a picture compared and input changes proposed.

Assistant in the top bar opens a side panel where you can ask questions about SOCA and catchment hydrology. Answers come from an AI service through the developer's server and are based on this manual: for questions about SOCA the assistant names the manual section it used, and when the manual does not cover something it says so.

- **It cannot see your screen.** A typed question sends only the question and the last few messages. **Explain my results** also sends a summary of this project's results (sub-basin names, areas, curve numbers, times of concentration, slopes, flags and the main settings); **Explain the log** sends the last lines of the log – never study-area coordinates, file or project names.
- **Pictures** – click 📎, paste a screenshot into the question box (Ctrl+V) or drop it on the panel, e.g. sub-basins from an HEC-HMS model or your own calculation, and ask why they differ. The picture is read first and the answer starts with *Values I read from the image* – check them before trusting the comparison. The results summary is sent with the picture so they can be compared; pictures are not kept.
- **Changing inputs** – ask, for example, "threshold 0.5 km² and fit the sub-basins to 2–10 km²" or "switch to large river mode". The assistant lists the proposed changes as *now* → *new* with **Apply** and **Dismiss**; nothing changes until you press Apply, and you still press **Run**. It can change the step 2 and 3 settings only (sub-basins, threshold, min/max area, fit to range, terrain data, DEM buffer, area cap) – never your study area, files or exports. ARC stays II.
- **3 questions per computer per day**; pressing Apply is not a question. The line under the box shows how many are left. The developer can change the limit, switch the assistant off, or turn it off for a computer.
- It needs the internet. Answers come from an AI and can be wrong – they are not engineering advice; check design decisions with a qualified engineer and your local standards.

3 Step 1 – Study area and outlets

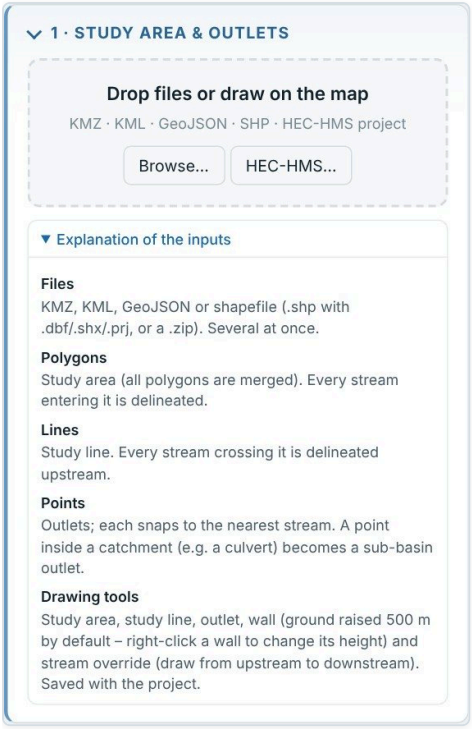


Figure 3.1 – Step 1 with its explanation open.

Drop one or several files on the box or on the map, or press **Browse...**. Accepted: **KMZ, KML, GeoJSON** and **shapefiles** (.shp with its .dbf / .shx / .prj, or a .zip holding them). Coordinates are read from the .prj when present; GeoJSON and KML are WGS 84.

Geometry	Meaning in SOCA
Polygon	Study area (several polygons are merged). Every stream entering it is delineated; the area inside is kept as local strips.
Line	Study line – e.g. a road centre line. Every stream crossing it is delineated upstream of the crossing. A line lying inside the catchment of another line further downstream adds a sub-basin outlet at every stream crossing (named *Line 2 crossing 1* ...), like an outlet point added by hand, kept even below the minimum area.
Point	Outlet – a culvert, dam or pond. It snaps to the nearest stream cell within 5 cells – onto a confluence if one is within 3 cells. A point inside a catchment becomes a sub-basin outlet.

HEC-HMS... opens an HEC-HMS project (.hms) – see 3.2.

3.1 Drawing tools

The toolbar on the left of the map draws the same inputs directly: **study area** (polygon), **study line**, **outlet**, and two DEM edits:

- **Wall** – raises the ground along the line (500 m by default; right-click the wall to change its height), e.g. an embankment or a bund the 30 m DEM does not show. The wall height becomes part of the terrain.
- **Stream override** – drawn from upstream to downstream, it forces the flow along the line (a channel, a culvert under a road, a diversion).

The **outlet** tool stays on after each point, so several points are added click by click; Esc (or the tool button again) stops it. Lines and polygons finish with a double-click or Enter.

Drawn items are listed under *Drawn on the map*, saved with the project, and can be edited, moved or deleted (right-click). A right-click on any drawn shape also offers *Use as* to change its role.

3.2 Opening an HEC-HMS project

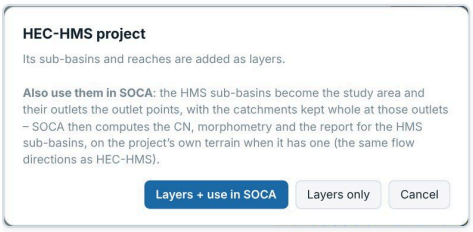


Figure 3.2 – Opening an .hms file.

Open or drop a **.hms** file. SOCA reads its basin models (sub-basin polygons, reaches and outlets from the HMS GIS files) and offers:

- **Layers + use in SOCA** – the HMS sub-basins are added as layers *and* become SOCA's inputs: their union is the study area, their outlets are the outlet points, and sub-basin division is switched off. SOCA then computes CN, morphometry and the report for the same sub-basins.
- **Layers only** – shown on the map for comparison.

Each sub-basin's outlet is taken from its own stream link (matched by the terrain watershed number). Where several links meet at one junction, the outlet is the link's last cell above the junction, and these imported outlets stay exactly on their stream cell. When the project has its own terrain, SOCA uses that DEM with its HEC-HMS method, so the flow directions – and therefore the sub-basins – are the same as in HEC-HMS (Appendix B).

4 Step 2 – Sub-basins

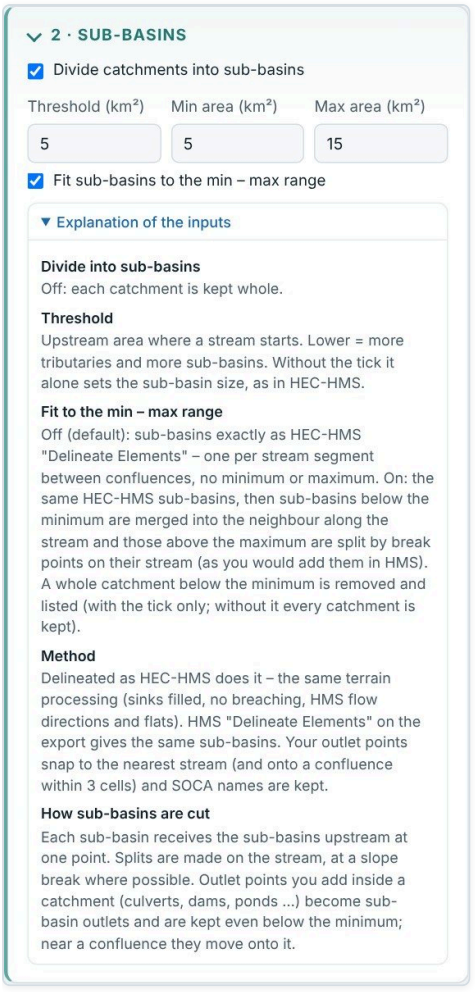


Figure 4.1 – Step 2 with its explanation open.

Input	Meaning
Divide catchments into sub-basins	Off: every catchment is kept whole (one unit per catchment).
Threshold (km²)	Upstream area where a stream starts. Lower = more tributaries and more sub-basins. Without the tick below it alone sets the sub-basin size, as in HEC-HMS.
Min / max area (km²)	Used with the tick below. With the tick, a whole catchment below the minimum is removed and listed in the results. Without it every catchment is kept; only sheet-flow strips inside a study area below the minimum are removed.
Fit sub-basins to the min – max range	Off (default): exactly the HEC-HMS sub-basins. On: the same sub-basins, then merged below the minimum and split above the maximum (4.2).

4.1 One sub-basin per stream segment (default)

SOCA delineates exactly as HEC-HMS *Delineate Elements* does R10: the same terrain processing (Appendix A.1), one sub-basin per stream segment between confluences, every catchment kept and no minimum / maximum area – raise the threshold for fewer, larger sub-basins. With the same DEM, threshold and outlet points, SOCA and HEC-HMS give the same filled DEM, the same flow directions on all but a handful of cells in 13 million, and the same sub-basins and reaches (Appendix B.2). Your outlet points snap as described in step 1 (Chapter 3) and the SOCA names are kept. When your own DEM is used with this option, the whole DEM is processed at once, as HEC-HMS does.

4.2 Fit to the min – max range

With the tick on, SOCA starts from exactly the HEC-HMS sub-basins (same terrain and flow directions) and adds one step: a sub-basin below the minimum is merged into its neighbour along the stream (a small headwater tributary joins the segment it flows into), and a sub-basin above the maximum gets break points on its main stream (an even split, slope breaks preferred) – as you would add break points in HMS Tool. A small piece that can only join a neighbour already near the maximum pushes it over; that sub-

basin is then split again on its main stream, both parts inside the range. A sub-basin still above the maximum is flagged with the reason. The flow directions are not changed, so HMS gives the same sub-basins from the exported streams and break points.

Naming

Catchments C-1, C-2 … are numbered by size, largest first; catchments whose streams leave through the same outlet share the number with letters (C-1-A, C-1-B).
Outlets are numbered from the catchment outlet *upstream*. With **one catchment** they are simply O-01, O-02 …: O-01 is the final outlet, O-02 the next one up. With **several catchments** the catchment number comes first, O-<catchment>-<n>: O-1-01 is the final outlet of C-1, O-2-01 that of C-2.
Sub-basins take the name of the outlet they drain to: B-02 (or B-1-02) drains to O-02 (O-1-02); when several drain to the same outlet (the branches above a confluence) they get letters, B-02-A, B-02-B.
Reaches R-03 (or R-1-03) run from outlet O-03 to the next outlet downstream (O-02).
The numbers grow with the count, for outlets and sub-basins alike: one digit up to 9 (O-1), two from 10 to 99 (O-01), three from 100 (O-001). Catchments of separate drainage systems continue the same numbering (C-1, C-2 …). Tool

5 Land use, soil and CN (automatic)

This part needs no input: it runs automatically after the sub-basins. Each cell receives a curve number from the **TR-55 table** R1 for its land-use class and hydrologic soil group:

- **Land use** – Esri / Impact Observatory 10 m annual land cover (Sentinel-2) R6.
- **Hydrologic soil group A–D** – HYSOGs250m, NASA ORNL DAAC R7. Dual groups (A/D, B/D, C/D) are taken as D (undrained), as in GCN250 R8.

Only your area is downloaded; no account is needed. Cells without a land-use class (no data, clouds) take the most common class of their sub-basin; the filled area is reported and flagged above 25 %.

SOCA uses the **average antecedent runoff condition (ARC II)** throughout – for the curve numbers of the report and the HEC-HMS losses, and for the SCS lag. ARC I (dry) and III (wet) values are given in the Excel tables for reference; the choice of a design condition belongs to the rainfall–runoff stage.

5.1 The CN table

The table is fixed (not user-editable) so that every project uses the same, checkable values. Built-up cells use the TR-55 *commercial and business* values (85 % impervious) – conservative for runoff.

Esri class	TR-55 cover used	A	B	C	D	Source
Water	Open water (impervious value)	98	98	98	98	Tool
Trees	Woods, fair	36	60	73	79	Table 2-2c
Flooded vegetation	(no TR-55 entry)	85	85	85	85	Tool
Crops	Row crops, straight row, good	67	78	85	89	Table 2-2b
Built area	Commercial and business, 85 % impervious	89	92	94	95	Table 2-2a
Bare ground	Fallow, bare soil	77	86	91	94	Table 2-2b
Snow / ice	as water (flagged)	98	98	98	98	Tool
Rangeland	Desert shrub, fair (arid rangeland)	55	72	81	86	Table 2-2d

Engineering judgement

In arid regions much of the land is mapped as *bare ground*, which TR-55 has no exact entry for; “fallow, bare soil” (CN 77–94) is the usual conservative choice. Where your project has a local CN guideline, or where gravel plains and desert pavements are known to infiltrate more, use the **CN raster override** (step 3) with your own values.

6 Step 3 – Advanced settings

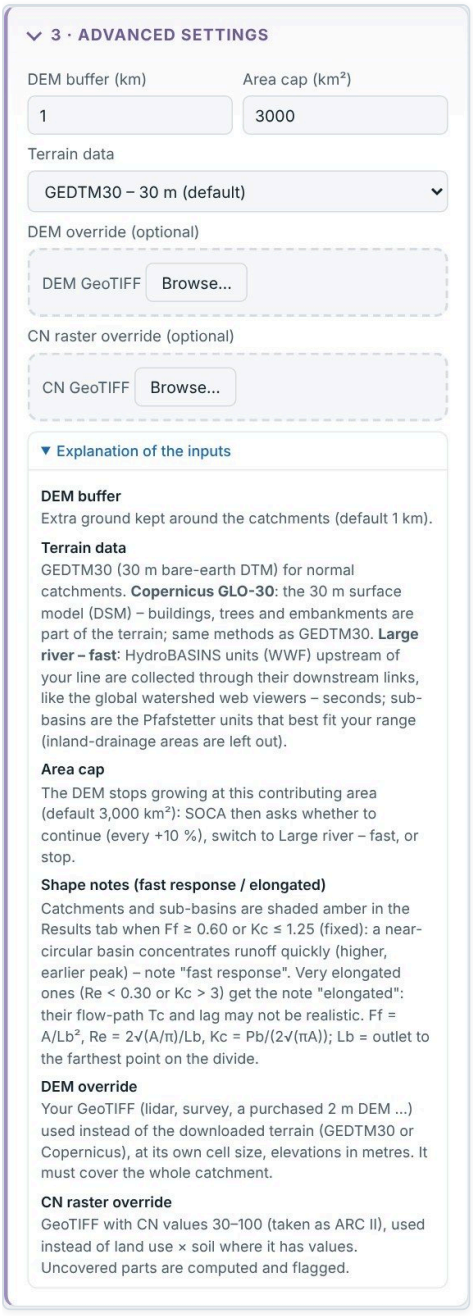


Figure 6.1 – Step 3 with its explanation open.

Input	Meaning
DEM buffer (km)	Extra ground kept around the catchments (default 1 km).
Area cap (km²)	Contributing area at which SOCA asks before growing the DEM further (default 3,000 km², Chapter 7).
Terrain data	**GEDTM30** – 30 m bare-earth DTM R5 (default); **Copernicus GLO-30** – 30 m surface model R20 (DSM: buildings, trees and embankments are part of the terrain), same methods; or **Large river – fast** (HydroBASINS, seconds) R11 – see 6.1, including the questions SOCA asks for large rivers.
DEM override	Your GeoTIFF (lidar, survey, a purchased 2 m DEM ...) used instead of the downloaded terrain (GEDTM30 or Copernicus), at its own cell size, elevations in metres. It must cover the whole catchment; otherwise the run is marked incomplete. A DEM in a projected CRS is used on its own grid, cell for cell.
CN raster override	GeoTIFF with CN values 30–100 used instead of land use × soil where it has values; uncovered parts are computed and flagged. Values are taken as ARC II.

6.1 Which terrain option?

Option	Use for	How it works
GEDTM30 (30 m)	Normal catchments up to a few thousand km²	30 m bare-earth DTM (buildings and trees removed), downloaded for your box only; the full SOCA delineation. If it cannot be reached, Copernicus GLO-30 is used and reported.

Copernicus GLO-30 (30 m)	Where you want the surface as the satellites see it (e.g. to compare with another study)	30 m surface model (DSM): buildings, trees and embankments stay in the terrain; the same SOCA delineation as GEDTM30.
Large river – fast	Very large rivers (e.g. the Nile above Cairo)	HydroBASINS level-12 units upstream of your input are collected through their downstream links (the continent's HydroBASINS file, about 120 MB, is downloaded once and kept with the project's downloads); slopes and flow paths come from GEDTM30 read for the basin only, at a coarse cell size; sub-basins are the Pfafstetter units that best fit your range; inland-drainage units are left out. Exact to within one unit (~100–200 km²).

SOCA asks before anything is downloaded:

- **Large river with GEDTM30** – a map of large rivers (HydroRIVERS, rivers draining ≥ 1,000 km², whole world; only the 10° tiles around your inputs are downloaded, a few hundred KB; offline the check is skipped with a note) R11 shows that your line, area or an outlet sits on a river bigger than the area cap: **Large river – fast**, **GEDTM30 – 30 m** (side catchments delineated, the large river itself incomplete) or **Cancel**. 30 m terrain alone can miss such rivers on flat, built-up valley floors.
- **Estuary** (Large river – fast) – HydroBASINS ends a big river where it meets the sea, so a line further out in a wide estuary (e.g. the Amazon) does not touch it; when the mouth of a river bigger than everything the line crosses lies within 50 km: include that river's whole basin, **Yes** / **No**.
- **No large river** (Large river – fast) – no river draining more than the area cap is crossed or within 50 km: **Large river – fast** (carry on), **GEDTM30 – 30 m** (start again at 30 m; the sub-basin values go back to 5 / 5 / 50 km² only if they are still the river defaults) or **Cancel**.

7 Running – progress, area cap and offline

Press **Run**. The progress bar shows the six steps. Where the amount of work is known (DEM tiles, land-use blocks, sub-basins) the bar shows real progress; otherwise a striped bar with the elapsed time and the current action. Results appear on the map as each step finishes. **Stop** ends the run; nothing already written is lost.

7.1 How the catchments are completed

SOCA starts with a DEM box around your inputs plus the buffer, delineates, and checks whether any catchment reaches the box edge. If it does, the box grows on those sides only and the delineation is repeated, until no catchment touches the edge. Each pass grows the box by at most 2.5 times its area, so no pass reads far more terrain than needed Tool. For very large boxes the cell size is coarsened so the grid stays within about 60 million cells.

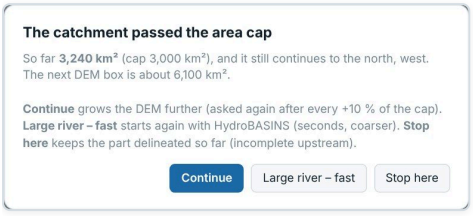


Figure 7.1 – The area-cap question.

7.2 Area cap

When the contributing area passes the cap, SOCA asks:

- **Continue** – grow the DEM further; asked again after every +10 % of the cap.
- **Large river – fast** – start again with HydroBASINS (seconds, coarser).
- **Stop here** – keep the part delineated so far (marked incomplete upstream).

If the next box would need more memory than the PC has free, growth stops with a warning.

7.3 Offline

SOCA needs the internet to start (Chapter 2). If the connection drops during a run, SOCA runs what it can (your DEM and CN raster are used). The run stops at the first step that needs a download; the steps done so far are kept, and **Complete now** continues from that step once you are back online.

Some data (the soil groups and the large-river map) comes from SOCA's public copy on GitHub. A connection that does not answer is tried **three times** and then once more by a second route before SOCA stops; the message then reads *Download problem – GitHub could not be reached*, because other sites may still work. Press Complete now to try again.

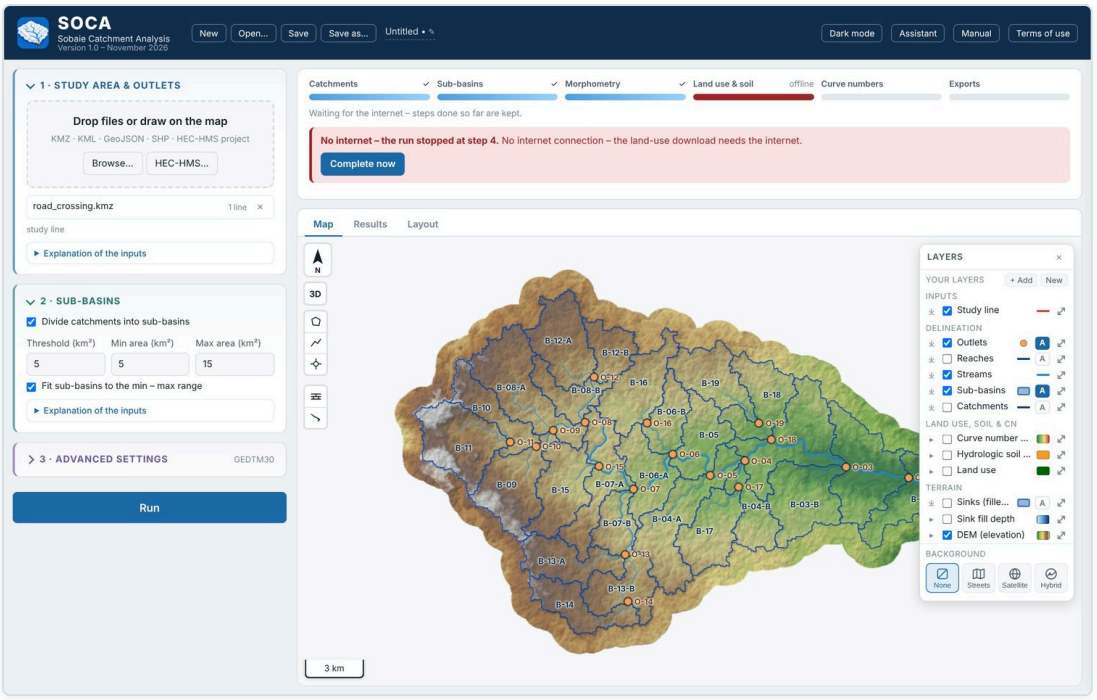


Figure 7.2 – A run waiting for the internet.

7.4 The log

Every run writes **SOCA log.txt** (each step, the data used, timings and warnings). **Export ▾ → Save log...** saves it any time, including the lines added after the run (satellite image and report figures), which is useful when reporting a problem.

8 Map tab – layers and labels

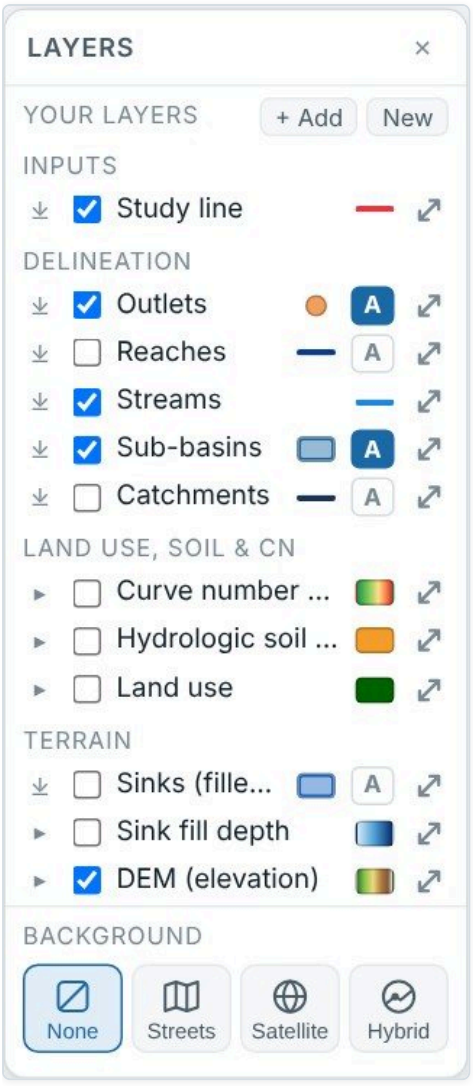


Figure 8.1 – The layers panel.

Layers are grouped: **Your layers**, **Inputs**, **Delineation** (outlets, reaches, streams, sub-basins, catchments), **Land use, soil & CN** and **Terrain**. Each row has:

- the tick (show / hide) and the name;
- the **style swatch** – click to change colour, line width, fill and transparency;
- the **A** button on layers with names – labels on / off and their style;
- ⤢ – zoom to the layer.
- ▸ on rasters opens the class or colour legend and the transparency slider; on your own layers it opens the draw tools.
- ⤓ on SOCA's vector layers (outlets, reaches, streams, sub-basins, catchments, sinks, study area, line ...) exports that layer alone – Shapefile, KMZ, GeoJSON or DXF, in your export coordinate system, with the attributes of its map popup.

Click a feature to see its values; Esc or a click on an empty part of the map clears every selection (SOCA features, your layers, drawn items, a Ctrl + click set).

Background: none, streets, satellite or hybrid. **3D** (under the north arrow) tilts the terrain; the exaggeration slider appears while 3D is on.

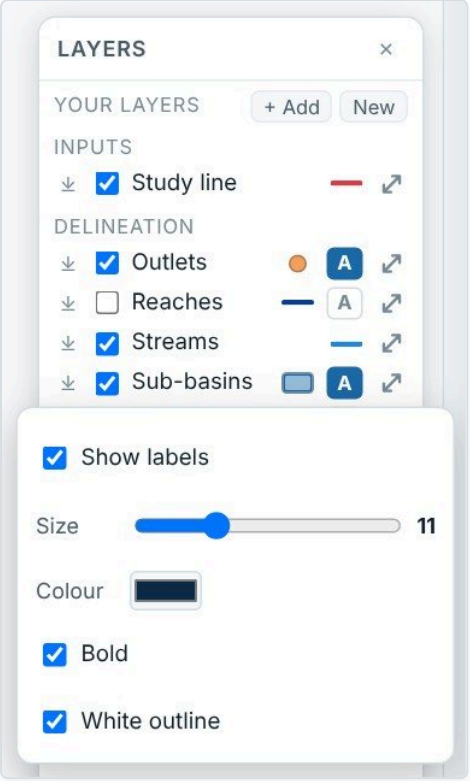


Figure 8.2 – The label menu (A).

8.1 Labels

The **A** menu sets, per layer: *Show labels*, text size, colour, bold and a white outline. A filled A means labels are on. What is labelled on the map is labelled in the layouts and their exports; hidden layers and layers without labels never print. Overlapping labels are skipped.

8.2 Sinks

Under **Terrain**: **Sink fill depth** is a raster of filled DEM − original DEM (metres per cell, walls included) – where and how much the terrain was raised before the flow directions (A.1); the fill volume of any area is the sum of its cells × the cell area. **Sinks** shows one polygon per filled depression with its area, fill volume, maximum depth, spill elevation and sub-basin (click it). The ▸ of Sink fill depth sets which sinks are drawn – minimum depth (default 0.1 m) and minimum cells (default 3) – to hide DEM noise; the raster keeps every cell. The totals are in the log and the report; the raster is in Export ▾ → Rasters.

8.3 Data for one polygon

Right-click any polygon – a sub-basin, a catchment, the study area or a polygon of your layers – and choose **Download for this polygon**: **DEM** (GEDTM30, 30 m), **Land use** (Esri 10 m), **Soil group** (HYSOGs250m) or **Curve number** (land use × soil with the CN table of 5.1, ARC II). The data is fetched for that polygon only – no run needed – clipped exactly to it (outside = no data), saved as a GeoTIFF where you choose (in your export coordinate system; land use and soil with a legend CSV) and added to **Your layers** with its legend. Polygons over 5,000 km² ask first.

8.4 Your layers

+ Add brings in your own GIS files (KMZ, GeoJSON, shapefile, DXF with a coordinate-system picker) for reference; **New** creates a layer to draw in. The ⋯ menu renames, exports (GeoJSON, KMZ, Shapefile or DXF, in UTM or WGS 84), orders or removes a layer. Your layers are saved with the project.

Click any feature for its attributes (for a sub-basin: area, CN, flow path, Tc, flags ...).

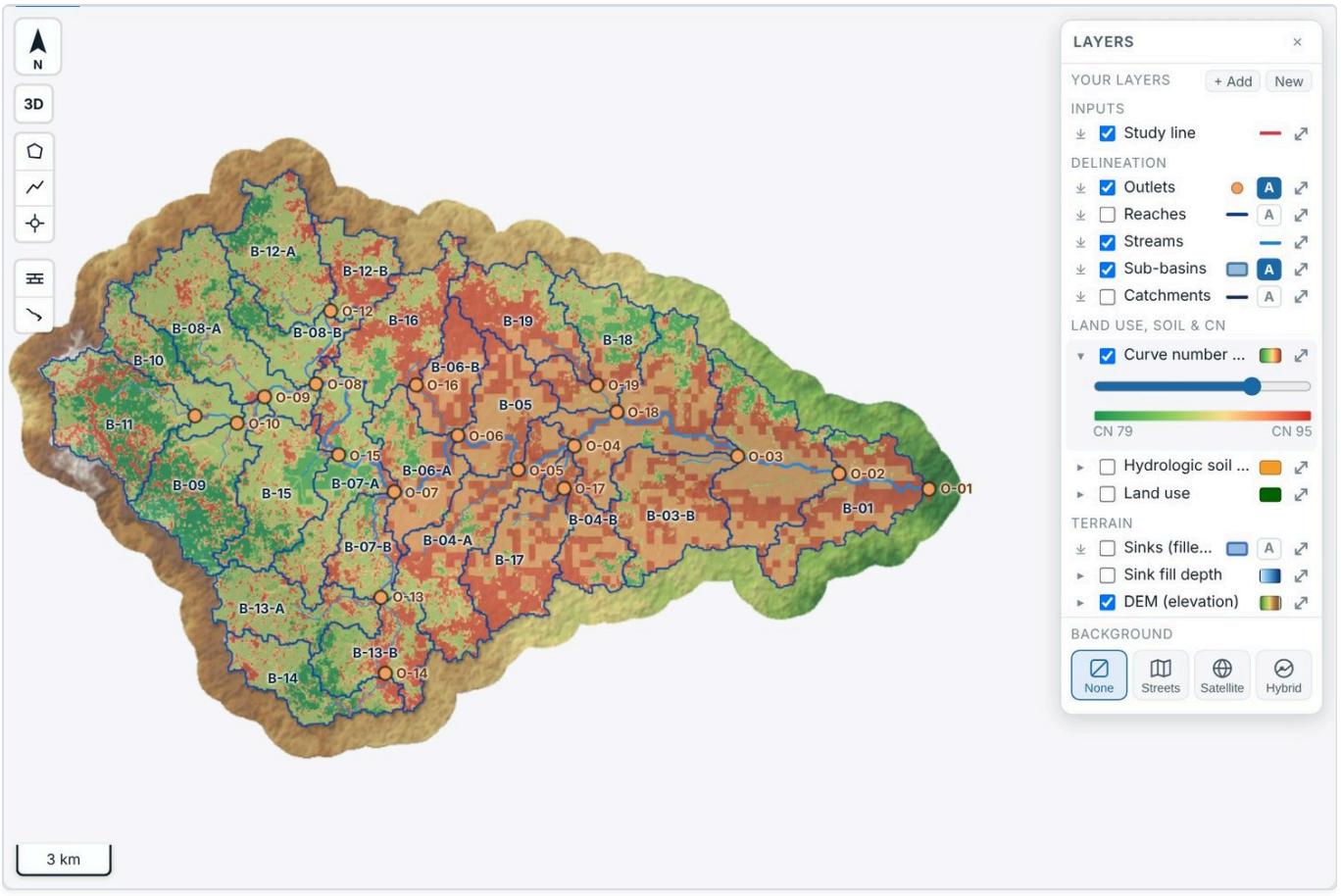


Figure 8.3 – The curve-number layer with its legend (synthetic land-use data).

9 Results tab

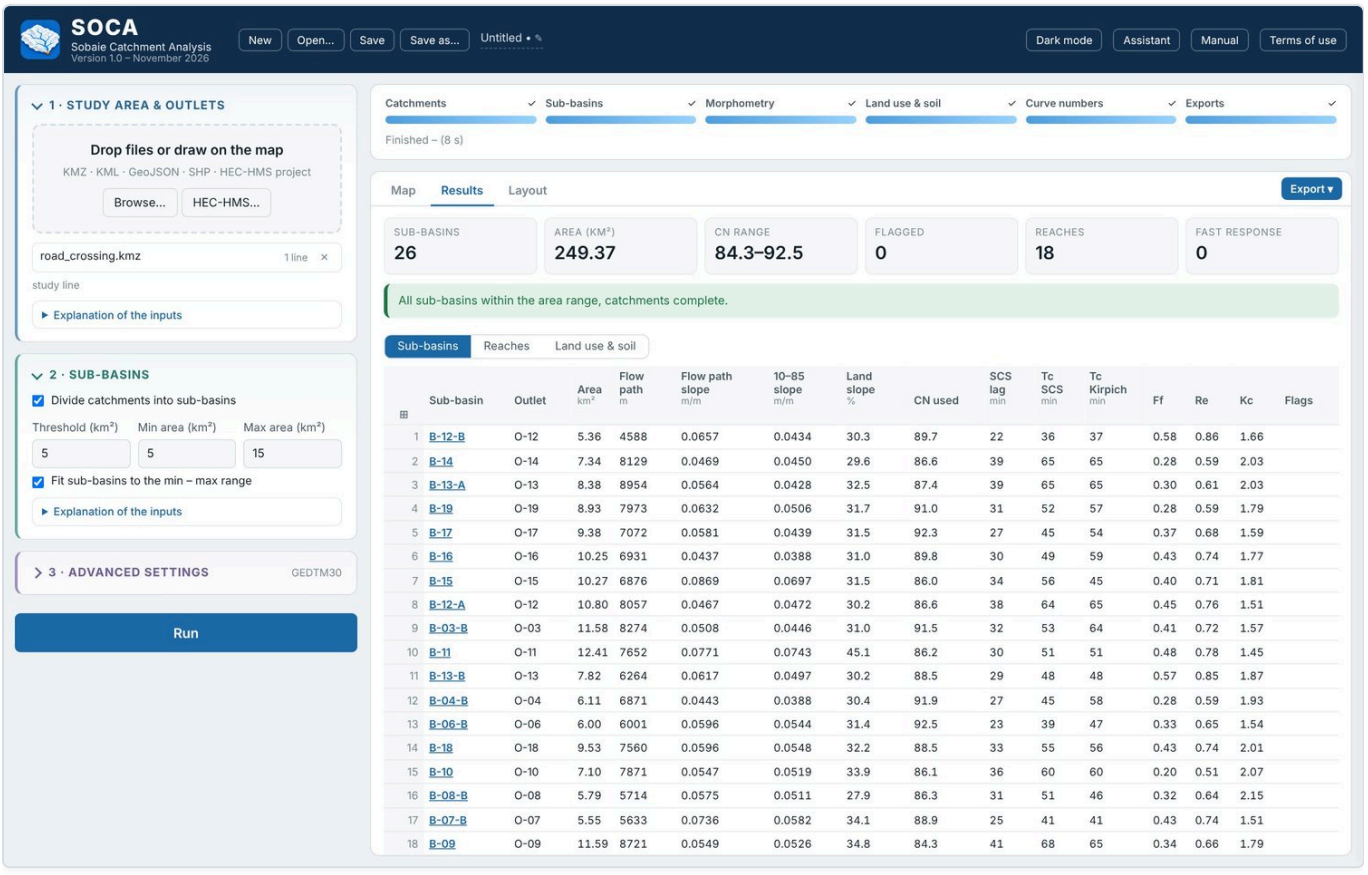


Figure 9.1 – Results: summary tiles, status line and the sub-basin table.

The tiles give the number of sub-basins, the total area, the CN range, flagged sub-basins, reaches and near-circular (“fast response”) sub-basins. The status line says whether all catchments are complete and lists removed small catchments. Three tables follow:

- **Sub-basins** – outlet, area, longest flow path and its slopes, land slope, CN used, Ff, Re, Kc and flags. Click a name to zoom to it on the map.
- **Reaches** – each reach from one outlet to the next: length, bed levels, drop, slope, 10–85 slope and Strahler order.
- **Land use & soil** – percentage of each land-use class and soil group per sub-basin.

Click a column header to sort; each table keeps its sort when you switch tabs, and the sorts are saved with the project (a new run starts unsorted); the ⊞ corner (or Ctrl + A) selects the whole table to copy into Excel. The full set of values – centroid, mean elevation, flow-path drop, main reach, SCS lag and T_C, Kirpich T_C (comparison), CN for each ARC, land-use and soil percentages – is in the Excel tables and the report.

9.1 Flags

Flags are kept short; red ones are problems, amber dashed ones are notes. The same texts appear on the map, in the Results tab, the report, Excel and the GIS attributes.

Flag	Meaning

fast response (note)	Near-circular shape: Ff ≥ 0.60 or Kc ≤ 1.25 (fixed limits) – runoff concentrates quickly (higher, earlier peak). Shown, not changed.
elongated (note)	Very long and narrow: Re < 0.30 or Kc > 3 – the Tc and lag from the longest flow path may not be realistic; check the shape or merge it with a neighbour.
above max	Fit to range: still above the maximum area – no split point on its main stream gives two parts inside the range (often after a small piece below the minimum could only join this sub-basin).
below min (P1) / below min (Line 1) (note)	Smaller than the minimum but kept: it drains to one of your outlet points, or to a stream crossing of an upstream study line.
catchment merged in (note)	A small neighbouring catchment (a separate stream crossing the line) was joined to this one; the main outlet is kept.
merged by you (note)	You merged these sub-basins on the map (Ctrl + click, right-click › Merge).
lag CN 50 / lag CN 95	The sub-basin CN is outside 50–95; the lag equation uses the nearest limit, as NEH-630 requires R3.
lag area > 49 km² (note)	The SCS lag equation was developed on 24 watersheds of 1.3 acres to 9.2 mi² (most under 2,000 acres); Folmar and Miller (2008) found that a reasonable upper limit may be as much as 19 mi² (≈ 49 km²) R3 R21. Above it the lag is indicative – compare with other Tc methods.
lag estimated (note)	No usable flow path (a single cell or a flat sliver): lag = 0.6 × Kirpich Tc, or 3.6 min.
land use filled n %	More than 25 % of the sub-basin had no land-use class (no data, clouds) and was filled from its most common class.
CN raster n % / no CN raster	Your CN raster covers only part of the sub-basin / none of it; the rest is land use × soil. The share is rounded down (99.9 %, never 100 %).
… present	A land-use class that needs a look is present (e.g. snow / ice, taken as water).

10 Layout tab – map sheets

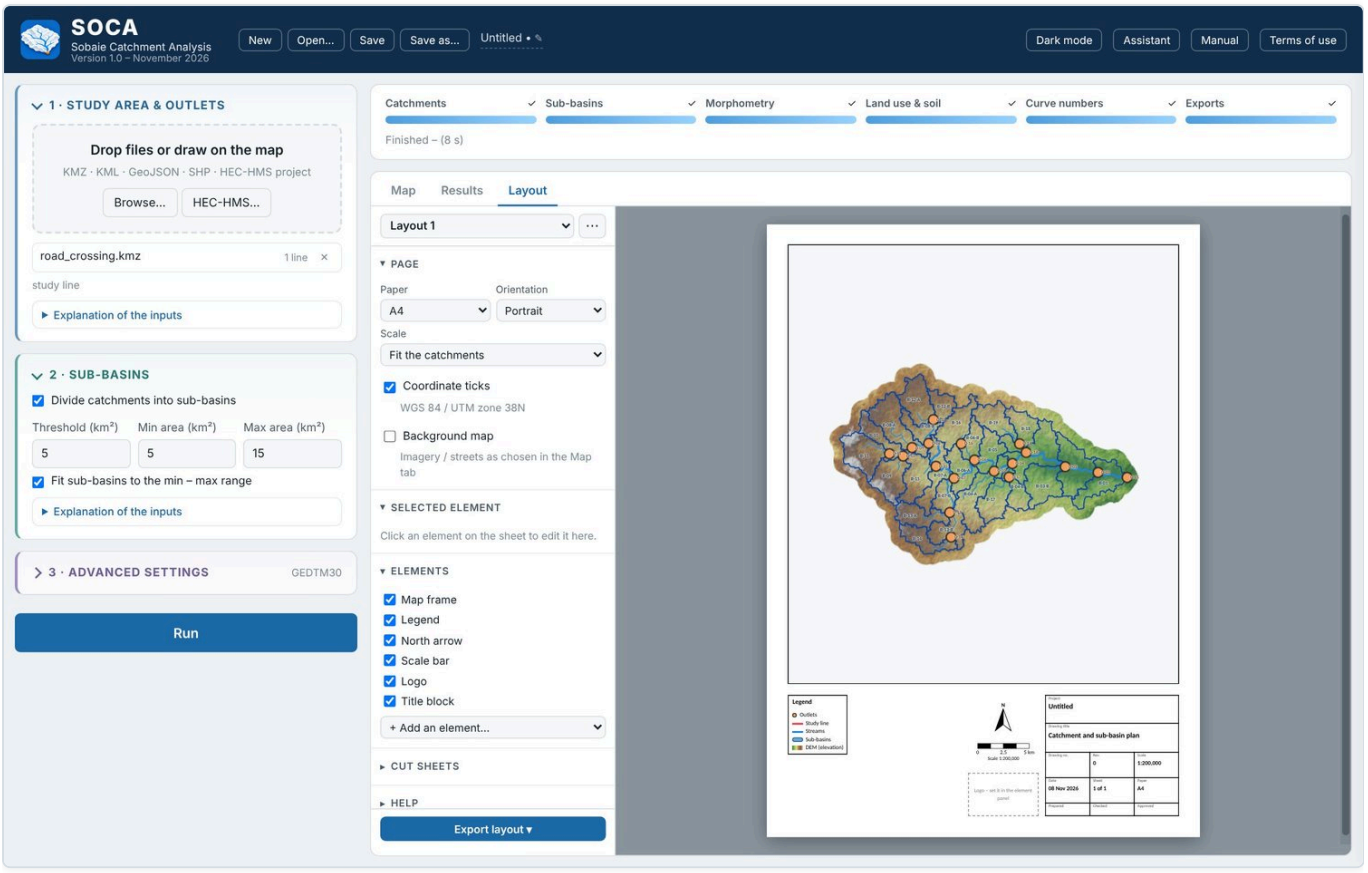


Figure 10.1 – The Layout tab: sheet preview and element options.

Layouts are printable map sheets (A4 to A0, portrait or landscape) with the map frame, title block, legend, north arrow, scale bar, coordinate ticks, logo, notes and a key plan. Several layouts can be kept in a project.

- **Elements**: click to select, drag to move, drag the handles to resize; Ctrl + click selects several. Right-click: copy, paste, duplicate, bring to front / send to back, hide, *Add to all layouts* (adds a copy, never replaces). Ctrl + C / V / D and Ctrl + Z (undo) work too.
- **Map frame**: dragging it moves the frame. Double-click inside it first (map mode) to move the map by dragging and zoom it with the wheel; click outside or press Esc to finish. Double-click the grey area to fit the sheet.
- **Cut sheets**: a long road or a large basin is split into sheets along the map (by scale or by number of sheets), with a key plan showing where each sheet lies.
  - **Key plan inset on every sheet** and **Match lines** (dashed lines with "Match line – see sheet N", off by default): ticks under Cut sheets.
  - **Key plan page**: the first page; it carries the key-plan inset too (no sheet highlighted), so every page has the same layout.
  - **Preview**: the box in the side panel – ◀, the list, ▶ (or ← / → with nothing selected) – shows the key plan and each sheet exactly as it will print.
  - **Adjust one sheet**: on a sheet, double-click the map to move or zoom that sheet alone; the key plan (and the match lines, when on) follow and the sheet is marked *adjusted*.

    - **Reset and undo**: **Reset this sheet** / **Reset all sheets** put them back; Ctrl + Z undoes. Adjustments are saved with the layout and reset when the scale, overlap, grid or paper changes.
- Text, legend and line sizes follow the paper (A4 → A0), and the legend skips empty layers. There is no page border: a map frame can fill the whole sheet.
- **Export layout ▾**: **PDF** (vector), **Image** – JPG (default, small) or PNG (lossless) at 150–400 dpi. Sheets are written one by one in the background with a progress bar, the elapsed time and Cancel.

11 Exports

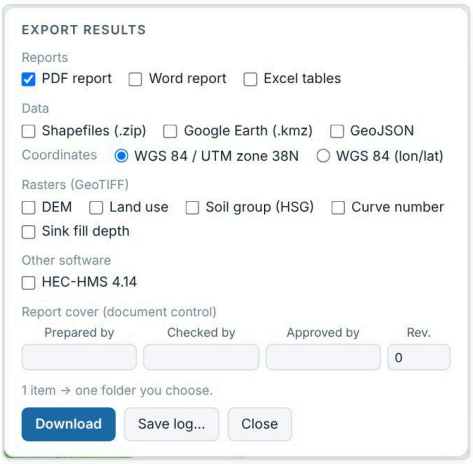


Figure 11.1 – Export ▾.

Group	Files
Reports	**PDF report** and **Word report** (identical content: inputs, data sources, methods, tables, flags, references; figures: location, Sentinel-2 image, shaded relief, land slope, catchments, sub-basins, land use, soil groups, curve number, HEC-HMS schematic; cover with document control – prepared, checked, approved, revision), **Excel tables**.
Data	Shapefiles (.zip), Google Earth (.kmz), GeoJSON – in project UTM or WGS 84. One layer alone: ⇩ in the layers panel (Chapter 8).
Rasters	GeoTIFF: DEM, land use, soil group, curve number (with legends), sink fill depth.
Other software	**HEC-HMS 4.14** project. Map sheets: Layout tab (Chapter 10).

Every file is re-opened after writing and area totals are cross-checked between tables. A file is never overwritten silently: when files with the same names are already in the folder, SOCA lists them and asks – **Replace**, **Keep both** (the new files get "(2)", "(3)" ...) or **Cancel**. The HEC-HMS project and the rasters always go into a new folder. Export is unavailable while a run is going; pressing New, Open or Run while an export is still going asks first – **Wait** or **Cancel the export**.

The report and the HEC-HMS export use the settings of the run that made the results; if a setting is changed afterwards, or a DEM or CN raster is added, replaced or removed, SOCA shows *Settings changed – press Run to update* (it clears when the setting or file is put back). A merge on the map re-runs from the sub-basins with the stream threshold, terrain, buffer, area cap, DEM and CN raster of the last run; new values are used when you press Run.

On every report map the legend sits inside the frame – bottom right, or the free corner where the terrain data ends – and the map is widened (never past the data) so that no legend, north arrow or scale bar covers the study area.

Fast reports. The report figures are drawn in the background right after a run, so exporting the report normally takes only seconds. Exporting while they are still being drawn simply finishes them first.

The satellite figure combines several low-cloud Sentinel-2 scenes; any part with no usable scene is filled from NASA's global true-colour image (NASA GIBS – Blue Marble Next Generation, about 500 m), and the caption says so. It is downloaded in the background during the run, for up to 5 minutes. When you export:

- if the image is ready, it is in the report;
- if it is still downloading, Export waits at most 1 minute and otherwise goes without it – the download continues, and a later export includes it automatically;

- for catchments over 20,000 km², and in *Large river – fast* mode, the figure uses NASA's global image instead (one download of a few seconds for any basin size; coarse, for orientation at that scale).

When the figure is missing, the report says why: too large an area, still downloading (export again in a few minutes), too slow a connection, or no connection.

While files are exported, a box at the bottom right shows the current stage (for example *Drawing the report figures 6 of 10*), a progress bar, the elapsed time and **Cancel**; it closes with *Saved … – 0:52*.

11.1 The HEC-HMS project

The export is a complete HEC-HMS 4.14 project in the same layout HMS itself writes R10: project, terrain (elevation, sink-filled DEM, flow direction, accumulation, streams), basin model with GIS geometry (sub-basins, reaches, outlets, flow paths), a meteorologic model, a control specification and a run. It opens, shows the map and computes.

- Each sub-basin: **SCS curve number** loss with the SOCA CN and 0 % impervious (TR-55 CNs already include impervious cover), **SCS unit hydrograph** with the SOCA SCS lag, no baseflow.
- Each sub-basin drains to a junction at its outlet; junctions are linked by reaches with **no routing** (Lag method, 0 min) – add routing (e.g. Muskingum-Cunge) where the reaches attenuate the flood.
- The storm is a **placeholder**: an SCS Type II 24-h hypothetical storm of 100 mm. Replace it with your design storm (for example from SORA), then compute Run 1.
- The time step is 1 minute; the simulation length covers the storm plus 5.5 × the largest lag.

12 Projects and privacy

12.1 Project folders

- **Folder**: every project is a folder, as in HEC-HMS – **Name\Name.soca** (the project file: settings, inputs, drawn items, your layers, layouts and table sorts), **inputs** (copies of your input files) and **data** (the run: DEM, flow grids, catchments, sub-basins, sinks, the land-use, soil-group and CN rasters, results and the log).
- **Save / Save as**: Save the first time asks where to create the folder; Run on a project not saved yet asks the same, then runs. Save as... copies the project, results included, to a new folder.
- **Move or share**: copy the project folder.
- **Open**: double-click the .soca file, or Open.... SOCA asks before closing, New or Open when changes are not saved. Saving does not change the results; *Inputs changed* appears only when an input really differs from the last run.
- **Defaults**: a new project (and SOCA when it starts) opens with the built-in settings – threshold 5, min 5, max 50 km², fit unticked, Esri 2025 land use, HYSOGs soil, area-weighted CN, ARC II (fixed), 1 km buffer, 3,000 km² cap, GEDTM30 – not the values of the last run; a saved project opens with its own.

12.2 SOCA's own folder and downloads

Terrain, land use, soil, the large-river map, HydroBASINS and the report's satellite image are downloaded into the project's **downloads** folder and reused by every run while the project is open (a re-run, or switching the terrain and back, takes seconds). When the project is closed – SOCA closed, New, or another project opened – only the files used by its latest run are kept, and the folder is held under 2 GB; the rasters cropped to the project stay in **data**. SOCA itself keeps only the unpacked program, its settings and temporary files in the hidden folder **%LOCALAPPDATA%\SOCA** – never projects or downloads; older program versions and temporary files are removed automatically, and so are the leftovers of earlier test versions (their caches in %LOCALAPPDATA%\SobaieLab and their temporary folders in the Windows temp folder); folders holding a project are never touched.

13 References

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Appendix A Methods and formulas

A.1 Terrain and streams

SOCA has one terrain engine for both sizing options, built on the rules HEC-HMS 4.14 uses (verified cell by cell, Appendix B):

- **Projection** – the DEM is used in the UTM zone of the site, or in your DEM's own projected grid cell for cell. With the default option, a DEM in latitude/longitude is reprojected as HEC-HMS does it: the GDAL default grid of the whole DEM, bilinear.
- **Fill** – exact Priority-Flood R19: every depression is filled to its spill level, perfectly flat, nothing lowered; the grid border and cells next to no-data are outlets.
- **Flow direction** – D8 R13: the steepest drop ÷ distance on the filled DEM, computed in single precision; neighbours are checked E, N, W, S, then NE, NW, SW, SE, and a tie goes to the first.
- **Flats** – Garbrecht–Martz R12: a gradient towards lower terrain plus a gradient away from higher terrain, added; where the two cancel, the procedure is repeated on the summed surface.
- **Streams** – cells whose upstream area (each cell counting itself) is at least the threshold; stream order is Strahler R14.

Walls raise the DEM along their line before the fill; stream overrides set the D8 direction along the drawn line.

A.2 Longest flow path and slopes

The flow distance from every cell to the outlet of its sub-basin is accumulated along the D8 directions (diagonal steps √2 × cell size). The longest flow path *L* starts at the most distant cell; its drop *H* is measured on the conditioned DEM. With the default option, the flow-path length, slopes, 10–85 slope and land slope follow the HEC-HMS definitions (raw DEM, path from the outlet's cell edge) so the two programs report the same values, and the full set of HEC-HMS sub-basin and reach characteristics is written into the HEC-HMS export.

$$S = H / L \ \text{(minimum 0.0005)} \qquad S_{10–85} = (z_{85} - z_{10}) / (0.75\ L) \qquad \text{R18}$$

z_{10} and z_{85} are the elevations on the path at 10 % and 85 % of *L* from the outlet. The mean land slope *Y* is the average over the sub-basin of the cell slope from central differences on the original (not conditioned) DEM, in %.

A.3 Time of concentration and lag

$$\text{Kirpich (metric):} \quad T_c\ [\text{min}] = 0.0195 \cdot L^{0.77} \cdot S^{-0.385} \quad (L \text{ in m, } S \text{ in m/m}) \qquad \text{R9}$$

$$\text{SCS lag:} \quad \text{Lag}\ [\text{h}] = l^{0.8} \cdot (S' + 1)^{0.7} / (1900 \cdot Y^{0.5}), \quad S' = 1000/CN - 10, \quad T_c = \text{Lag} / 0.6 \qquad \text{R3 eq. 15-4}$$

l is the flow length in feet, *Y* the mean land slope in %. The CN used in the lag equation is limited to 50–95, as NEH-630 Ch. 15 requires R3. The equation was developed on 24 watersheds of 1.3 acres to 9.2 mi² (most under 2,000 acres); Folmar and Miller found that a reasonable upper limit may be as much as 19

mi² (≈ 49 km²) R3 R21 – larger sub-basins get the note *lag area > 49 km²*. T_c = lag / 0.6. The HEC-HMS export uses the SCS lag, and the report's summary gives the SCS lag and T_c; Kirpich's equation (small, steep agricultural catchments in Tennessee) is shown for comparison.

A.4 Shape indices

$$F_f = A / L_b^2 \quad R_e = 2\sqrt{(A/\pi)} / L_b \quad K_c = P_b / (2\sqrt{(\pi A)})$$ R15 R16 R17

L_b = straight distance from the outlet to the farthest point on the divide; P_b = perimeter measured on the boundary simplified by one cell (the raster staircase would make it too long). Note *fast response* (amber) when $F_f \geq 0.60$ or $K_c \leq 1.25$ – $R_e = 1.128\sqrt{F_f}$, so $F_f \geq 0.60$ already means $R_e \geq 0.87$ and R_e is shown for information; note *elongated* when $R_e < 0.30$ or $K_c > 3$ Tool.

A.5 Curve number and ARC

CN per cell from the table in 5.1; the area-weighted CN is the cell average over the sub-basin R2. SOCA uses ARC II; the ARC I and III values listed in the Excel tables are converted from the composite ARC II CN of each sub-basin (not cell by cell), as NEH-630 Ch. 10 applies the antecedent condition to the drainage area R2 R4:

$$CN_I = 4.2 \cdot CN_{II} / (10 - 0.058 \cdot CN_{II}) \quad CN_{III} = 23 \cdot CN_{II} / (10 + 0.13 \cdot CN_{II})$$ R4

Appendix B Verification of SOCA

B.1 Comparison with HEC-HMS 4.14

Two DEMs of about 12,000 km² each were preprocessed and delineated in HEC-HMS 4.14 (threshold 10 km², one outlet each) and the same DEMs and outlets were run in SOCA with the default option (November 2026, SOCA 1.0):

Item	Project 1 (87 sub-basins, 43 reaches)	Project 2 (89 sub-basins, 44 reaches)
Reprojected DEM	same grid and identical values on every cell	
Filled DEM (Preprocess Sinks)	identical on every cell (largest difference 0.0 m)	
Flow directions	identical on all but 5 of 13,332,072 cells	identical on all but 1 of 13,108,160 cells
Flow accumulation	differs on 552 cells (downstream of those 5)	differs on 21 cells
Sub-basins	87 = 87; 86 on exactly the same cells, 1 overlapping 99.99 %	89 = 89; 88 on exactly the same cells, 1 overlapping 98.6 %
Reaches	43 = 43; lengths within one cell (30 m)	44 = 44; lengths within one cell (30 m)
Longest flow path, 10–85 length, basin relief, relief ratio	identical for 86 of 87 (largest difference 0.35 %)	identical for 88 of 89 (largest difference 0.17 %)
Centroidal length, 10–85 slope, elongation ratio	identical for 85–86 of 87 (largest difference 0.5 %)	identical for 85–88 of 89 (largest difference 0.6 %)
Longest and centroidal flow-path slope	identical for 54–56 of 87 (up to 2.7 %)	identical for 55–62 of 89 (up to 5.8 %)
Basin slope, drainage density	within 0.8 % and 0.002 %	within 3.6 % and 0.01 %

The few flow directions that differ are of two kinds: cells where two downhill neighbours are equally steep to the last digit of single precision (HEC-HMS's rounding picks the other one), and two cells in large flat areas (one by the sea at 0 m) that the flat-resolving step leaves without a direction. Each such cell moves the sub-basin boundary or the reach length by one cell where it lies. The flow-path slopes differ more because HEC-HMS sometimes takes the outlet elevation on the cell edge and sometimes in the outlet cell (largest in small sub-basins).

- **Opening an HEC-HMS project**: with the project's own terrain the same engine reproduces its flow directions (to the few cells in B.1), so the opened sub-basins match the project's.
- **Computation**: every build exports a project and computes it in HEC-HMS 4.14. Outlet areas and volumes balance.

B.2 Data and table checks

- The CN table was checked value by value against TR-55 Tables 2-2a–d.
- Every run checks that sub-basin areas add up to the catchment areas, that land-use, soil and CN tables cover the same area, and that every written file opens again.

Appendix C Limitations and good practice

- **Terrain resolution.** A 30 m DTM cannot see culverts, small channels, kerbs or embankments narrower than a few cells. In flat or urban areas check the streams against imagery; use walls, stream overrides or your own DEM where needed. Study areas under 1 km² are flagged.
- **Flat terrain and inland drainage.** Filling flats creates artificial parallel flow; large filled areas are reported in the log. Closed basins (sabkhas, playas) drain to sinks.
- **Global land use and soil.** 10 m land cover and 250 m soil groups are regional products: check them against site knowledge, especially bare ground in arid areas and the HSG of alluvial wadi beds. Use the CN raster override for project-specific values.
- **Tc methods.** The SCS lag and Kirpich equations have limited ranges (Appendix A.3); for large or urban sub-basins compare with the velocity (segmental) method.
- **HEC-HMS model.** It has no channel routing and a placeholder storm; add routing and the design storm before using its peaks.
- **Review.** SOCA results are engineering inputs: they must be reviewed by a qualified engineer before they are used for design.

Appendix D Glossary

Term	Meaning
ARC	Antecedent runoff condition – the soil wetness before the storm (I dry, II average, III wet).
Catchment	All the land draining to one point where a stream enters the study area, crosses the study line or reaches an outlet.
CN	Curve number (30–100) of the SCS / NRCS runoff method; higher = more runoff.
D8	Flow direction to the steepest of the eight neighbouring cells.
DTM / DEM	Digital terrain (bare-earth) / elevation model.
HSG	Hydrologic soil group A (high infiltration) to D (low).
Lag	Time from the centre of excess rainfall to the peak of the unit hydrograph (0.6 T_c in the SCS method).
Outlet	The downstream point of a catchment or sub-basin.
Reach	The stream between two outlets.
Stream threshold	Upstream area at which a cell is counted as a stream.
Study line	A line drawn or imported in step 1 (e.g. a road centre line); every stream crossing it is delineated upstream.
Sub-basin	Part of a catchment draining to one outlet; the unit of a rainfall–runoff model.
Cut sheet	One of several map sheets a large area is split into, with match lines to its neighbours.
Key plan	A small map of the whole area showing the sheet grid; the first page and an inset on every cut sheet.
Sink	A closed depression in the DEM, filled to its spill level before the flow directions are computed (Sink fill depth = filled − original DEM).
T_c	Time of concentration – travel time from the hydraulically most distant point to the outlet.

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